

Harnessing the power of language models in cybersecurity: A comprehensive review

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ABSTRACT

Language models are transforming cybersecurity by addressing critical challenges such as the growing skills gap, the need for expertise augmentation, and knowledge retention. These models offer scalable, adaptable, and round-the-clock defenses against evolving cyber threats. By generating human-like text, processing data efficiently, and providing actionable responses, language models bridge the gap between automated systems and human expertise for different cybersecurity applications. However, the application and adaptation of language models for cyber security is still in its infancy. This review explores the use of general models, such as BERT, and larger models in cybersecurity research. It provides a structured framework for developing customized language models tailored to applications including content analysis, software and systems analysis, threat intelligence and monitoring, and cyber vetting. The study critically examines challenges, such as data confidentiality, infrastructure requirements, integration complexity and the evolving threat landscape. Moreover, it underscores the need for transparency, responsible use, and bias mitigation to ensure reliable and secure deployment of these models. In addition, this work critically examines the socio-technical dimensions of language model integration, focusing on their impact on organizational workflows, decision making and human-machine collaboration. By considering both technical and socio-technical considerations, this review provides a roadmap for future research and development. It highlights the potential of language models to improve organizational resilience, ensure secure implementation, and support informed decision-making in cybersecurity practice.

1. Introduction

As cyber threats grow more sophisticated and pervasive, they jeopardize not only organizational resilience but also critical infrastructures and global security. The cybersecurity landscape is further complicated by the shortage of skilled professionals, an overwhelming amount of security data and challenges in knowledge retention due to workforce attrition. These factors create an urgent need for innovative, scalable and adaptable solutions to protect systems, data and digital infrastructures.

Large language models (LLMs), with their ability to process unstructured data, recognize patterns and generate context-aware insights, have emerged as transformative tools to address these challenges. These models can enable the automation of repetitive tasks, streamline incident response, and enhance threat intelligence, so that security analysts can focus on high-priority threats. Open source LLMs such as LLaMA and Falcon can further enhance this potential by enabling localized implementations that comply with regulatory requirements such as GDPR and

ensure sensitive data remains on-premises. Additionally, these models can provide cost-effective solutions by eliminating license fees and supporting offline deployment for air-gapped environments, which is critical for sectors such as defense and finance.

Despite their promising potential, the practical integration of LLMs into cybersecurity operations has not yet been sufficiently explored. Effective use requires the proactive adaptation of existing systems and processes to domain-specific demands (Bayer, Kuehn, Shanehsaz & Reuter, 2022). This includes fine-tuning models for customized applications, ensuring robust privacy and security measures, and complying with complex regulatory frameworks (Brown, Lee, Mirehshgallah, Shokri & Tramèr, 2022). While recent studies (Motlagh et al., 2024) have demonstrated the applicability of LLMs in different cybersecurity tasks such as detecting threats (defensive) and simulating attack scenarios (offensive), significant challenges remain. These include ensuring data privacy, adapting models to specific domains and achieving consistent operational effectiveness.

This paper aims to address this gap by evaluating the state-of-the-art

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language model applications in cybersecurity, identifying key use cases, and offering actionable insights for researchers, cybersecurity professionals, and organizations. Specifically, this review focuses on:

- An in-depth evaluation of the state-of-art applications of language models in cybersecurity domain.
- A structured approach for implementation of customized language models for cybersecurity by detailed discussion of specific dimensions and multi-faceted customization process.
- A comprehensive examination of challenges associated with the implementation of language models and also discusses the strategies that can be used to mitigate them.

By consolidating these dimensions, this study provides a structured framework for understanding how LLMs can be effectively customized and deployed to address real-world cybersecurity challenges. This review synthesizes qualitative and quantitative studies as well as real-world case studies to provide a holistic overview of the field, with inclusion criteria based on relevance, methodological rigor, and significance of contributions. Ultimately, this work aims to bridge the gap between theoretical advances and practical implementations, paving the way for customized, effective and privacy-conscious cybersecurity solutions.

The rest of the article is structured as follows. [Section 2](#) provides the relevant background information related to language models, exploring their potential and foundational concepts within the cybersecurity domain. In [Section 3](#), a detailed discussion of different fine-tuning methods from the literature for the development of customized models. [Section 4](#) offers the state-of-the-art literature review on the application of language models in cybersecurity domain, highlighting key advancements and use cases. In [Section 5](#), we discuss the distinct dimensions related to the structure of customized language models, focusing on their solicitation in the cybersecurity domain and detailing how these models can be tailored for specific cybersecurity needs. [Section 6](#) provides a thorough examination of both the technical and socio-technical challenges, alongside corresponding considerations, to address the complexities of implementation of customized language models in cybersecurity. Finally, in [Section 7](#), we conclude this study by summarizing the main findings and discussing future research directions.

2. Language models and their capabilities

Language models represent a formidable technology capable of generating data or content that closely mimics human-generated material. These models, available in variety of architectural forms, have the remarkable ability to understand patterns and structures in existing data, allowing them to generate new data that is both coherent and contextually relevant ([Ray, 2023](#)). Their wide-ranging applications in natural language processing (NLP), have generated immense interest in their transformative potential in a variety of domains, including the important area of cybersecurity.

Most language models use transformer architectures with attention mechanisms, while a few, like RWKV and Mamba, use different architecture. Transformer-based architectures serve as a foundation, with variants developed for different purposes. For instance, the bidirectional encoder representations from transformers (BERT) model are highly suited for natural language understanding tasks including text classification and sentiment analysis, due to its bidirectional understanding capability ([Devlin, Chang, Lee & Toutanova, 2019](#)) and smaller size, which makes it easier to train and deploy. On the other hand, generative pre-trained transformer (GPT) models are capable of generating and completing text, which is an essential for chatbots and content creation ([Ray, 2023](#)). These generative models have extraordinary adaptability and have applications beyond simple classification due to their ability to recognize and apply patterns from a wide variety of data formats.

The cybersecurity domain presents numerous research challenges

that require expertise in text analysis, information extraction, pattern recognition, real-time data analysis, contextual understanding, and natural language generation ([Zhou, Liu, Zhong & Zhao, 2021](#)). Language models are characterized by their exceptional contextual understanding, aggregation of insights from heterogenous data sources, and ability to recognize complex patterns in big data. These attributes along with their ability to communicate in understandable format makes them suitable for different cybersecurity tasks including threat intelligence and monitoring, content analysis, software and system analysis, and cyber vetting. This automation increases efficiency and provides powerful tools to proactively defend against evolving threats in complex digital environment. Although language models are pre-trained, they can be fine-tuned for specific tasks. Also, research studies have shown the incapability of pre-trained models when used directly for domain-specific tasks ([Bajaj & Edwards, 2023](#); [Bayer et al., 2022](#); [Garza, Hemberg, Moskal & O'Reilly, 2023](#); [Scanlon, Breiteringer, Hargreaves, Hilgert & Sheppard, 2023](#)).

3. Fine-tuning of language models

Fine-tuning is a critical process in the application of language models for the cybersecurity domain as this domain encompasses specialized terminology, unique data formats, and specific context understanding. Fine-tuning involves tailoring the pre-trained language models to understand and interact with cybersecurity specific data to deeply attune them to the nuances of the cybersecurity. It bridges the gap between generic pre-trained language models and the unique requirements of specific cybersecurity applications to ensure that the language model aligns closely with the security team's expectations.

Fine-tuning techniques for language models encompasses three main approaches: parameter adjustment methods, data-centric methods, and architecture extension methods, as shown in [Fig. 1](#).

3.1. Parameter adjustment methods

Parameter adjustment-based fine-tuning refers to a set of techniques used to tailor pre-trained language models for specific tasks or domains by directly modifying their internal parameters. These internal parameters involve altering the weights and biases within the model itself. There are mainly three parameter adjustment techniques: full model, adapter-based, and reparameterization based fine-tuning.

The full model fine-tuning is the most comprehensive form of parameter adjustment, where essentially all the parameters of language models are updated. We have used the BERT model for the graphical representation of different fine-tuning techniques, but the insights and methodologies discussed here can be extended to large language models, such as GPT. The full model fine-tuning process for the BERT model, as illustrated in [Fig. 2](#), involves step-by-step adjustment of the model's pre-trained parameters to customize it for a specific task. The figure highlights the sequential phases, starting with the use of task-specific input data, followed by fine-tuning of the previously learned weights, biases, layer parameters, learning rate, and other hyperparameters to better match the nuances of the new task. This process culminates in the task-specific output layer, which ensures that the final predictions are highly specialized for the intended application. This extensive fine-tuning is crucial when dealing with large datasets, where maximizing accuracy is crucial.

In contrast, reparameterization-based fine-tuning technique involves alteration of model's behavior by application of mathematical transformations on model parameters. In this process, new parameters can be added and adjusted to regulate the transformations. In this direction, researchers ([Hu et al., 2021](#)) proposed a new method for low rank adaptation (LoRA), which includes freezing the pretrained weights and creating low rank adaptations of attention matrices of the query and key layers as shown in [Fig. 3](#). In this technique, the core layers of the pre-trained model are frozen, i.e., their weights remain unchanged

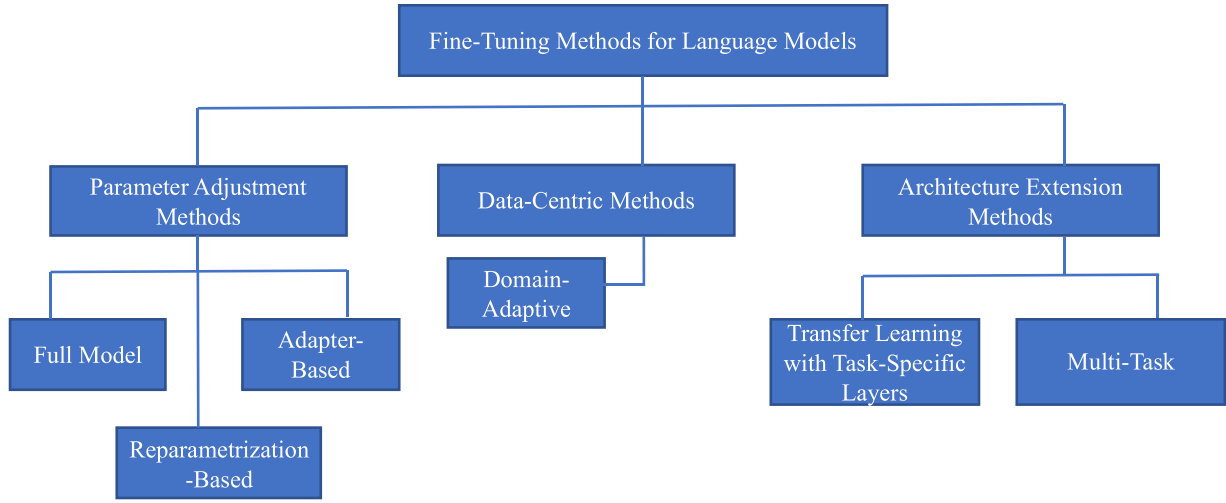


Fig. 1. Fine-tuning methods and techniques for customized language models.

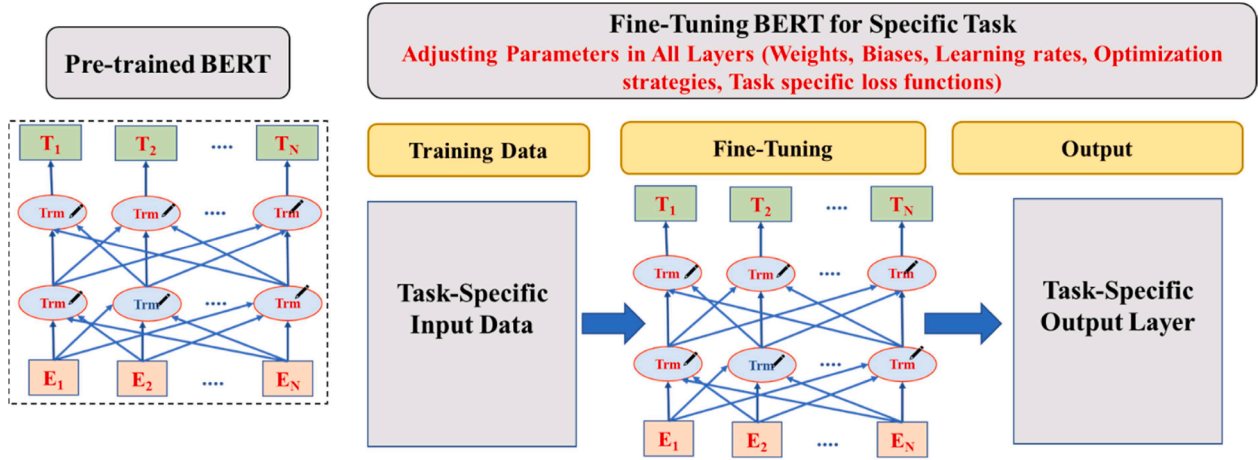


Fig. 2. Full model fine-tuning.

during the fine-tuning process. To adapt the model to a new task, low-rank adaptation (LoRA) modules are introduced into the attention mechanism of the Transformer architecture. These modules adapt the query and key metrics by adding some additional parameters, represented as low-rank layers, which change the behavior of the model without altering its primary structure. This model is particularly beneficial for resource-constrained environments as it significantly reduces the computational overhead while maintaining high performance. The figure clearly shows the separation between the frozen pre-trained components and the adaptable low-rank layers, highlighting how reparameterization enables effective and efficient task-specific fine-tuning.

Adapter-based fine-tuning involves insertion of strategically small, trainable layers (called adapters) into the model's existing architecture (Ding et al., 2023; Hu et al., 2023). These adapters consist of a down-projection layer, a non-linear activation function, and an up-projection layer as illustrated in Fig. 4. The down-projection layer reduces the dimensionality of the input features, the non-linear activation applies the transformation, and up-projection layer restores the features to their original dimensionality. For fine-tuning, these adapters are trained while the main transformer layers remain frozen, conserving computational resources and preventing overfitting. The input embeddings are processed by both the frozen transformer layers and the adapters, allowing the model to efficiently generate task-specific predictions. This approach is especially useful in resource-constrained

environments or scenarios that require quick and efficient adaptation of the new tasks.

3.2. Data-Centric methods

Data-centric fine-tuning concentrates on optimizing or adjusting the input data, rather than changing the architecture or parameters of the model. Thus, the main focus is on the careful selection and design of training dataset that prioritizes representativeness, reliability, and relevance to the target domain. The key technique in this method is domain-adaptive fine-tuning.

Domain-adaptive fine-tuning technique is suitable to ensure domain knowledge and context-specific understanding by focusing on preparation of customized dataset for particular domain, such as medical, legal or technical domain. While data selection is the central aspect of domain adaptation, secondary adaptations can also be made to the model's hyperparameters or training strategies to facilitate the fine-tuning process. However, these adjustments are usually aimed at optimizing the adaptation of the model rather than fundamentally changing its architecture or core parameters. Domain-adaptive fine-tuning is particularly valuable in scenarios where context-specific expertise and terminology are essential for accurate language understanding and generation within a specific domain.

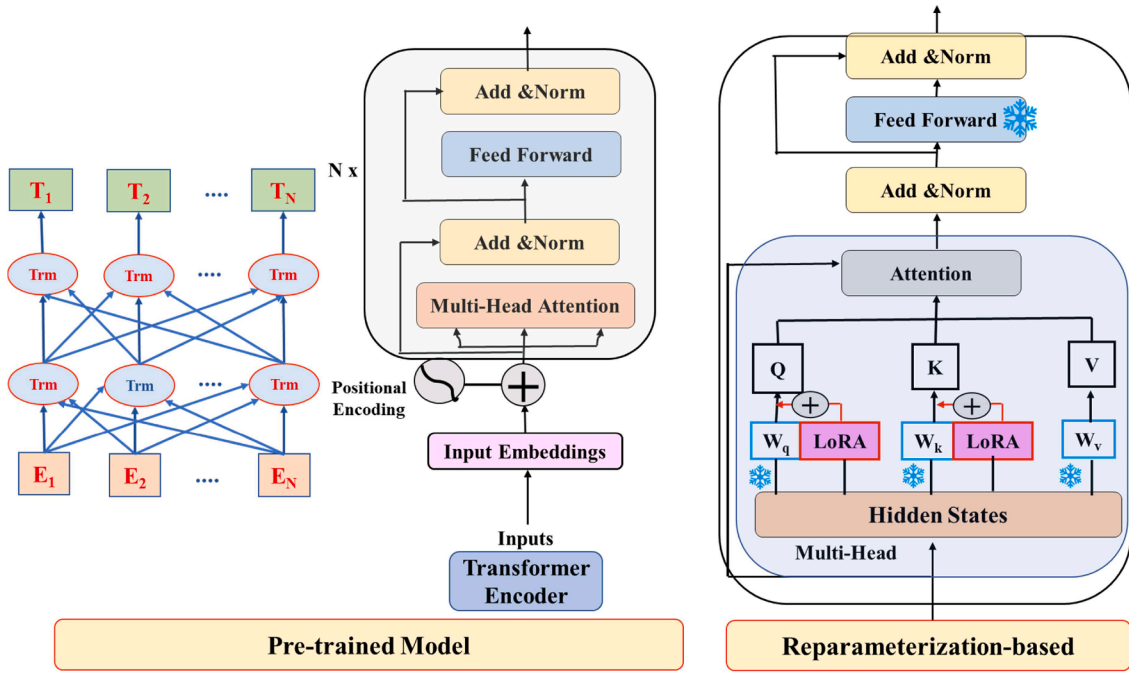


Fig. 3. Reparameterization-based fine-tuning.

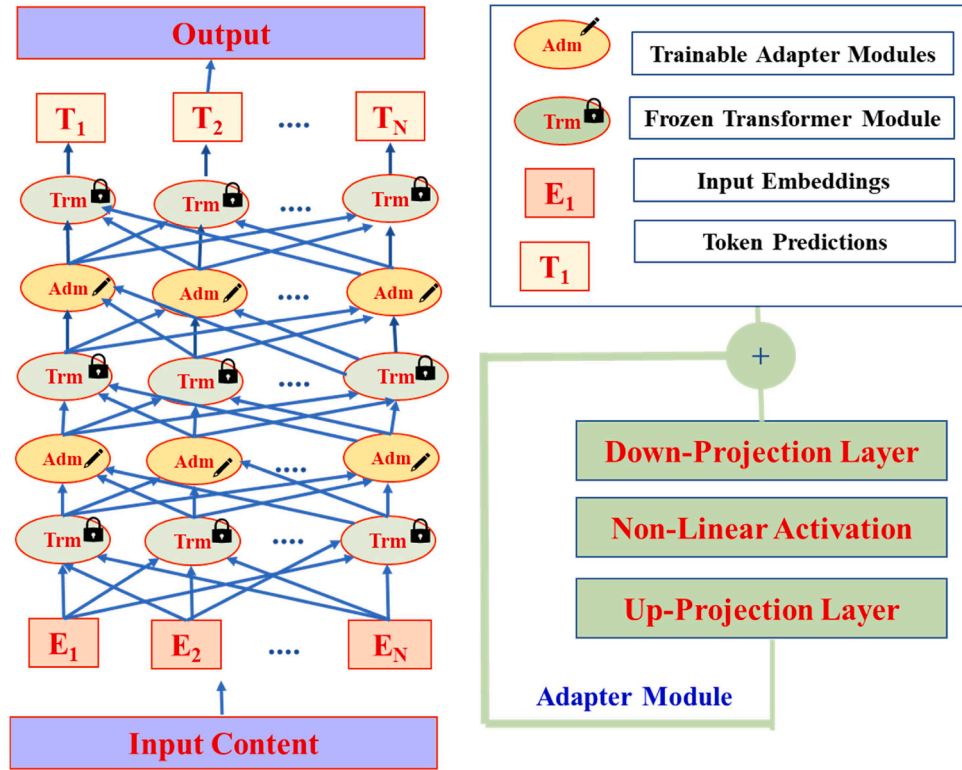


Fig. 4. Adapter-based fine-tuning.

3.3. Architecture extension methods

Architecture extension-based fine-tuning is an advanced method for customizing pre-trained language models by adding new structural components into their existing architectures (Thapa et al., 2022). For some applications (shown in Table 2), this method has been proven to provide significant performance improvements, but it also adds

complexity and might need more computing resources. There are two main techniques in this method: transfer learning with task-specific layers and multi-task learning. Although they both add new components to the architecture of language models, their functions and goals are different.

Transfer learning with task-specific layers involves the addition of new layers to the existing model architecture to improve its performance

on specific tasks. As shown in Fig. 5, these task-specific layers (highlighted in different colors) are added on top of the pre-trained model. The pre-trained model, which has already learned general features from a large dataset, serves as the foundation. The task-specific layers are designed to capture the unique characteristics of the new task, such as recognizing specific patterns or processing advanced data features. The diagram shows the flow of data through the task-specific layers, which are initialized with new parameters and trained using the task-specific data set. This process can involve fine-tuning the entire model or just the task-specific layers. The performance of the model is optimized by using both the generalized knowledge from the base model and the specific knowledge from the new layers. This multi-layered approach is crucial to adapt the model to the new task while maintaining the benefits of transfer learning.

On the other hand, multi-task learning based fine-tuning technique involves training a single model for multiple tasks simultaneously to enhance the adaptability and generalization capabilities of customized language model (Chaudhry et al., 2022). This technique is suitable for related tasks, as it uses diverse and balanced datasets to train the model. As shown in Fig. 6, the model architecture has shared base model (blue section) that extracts universal features applicable to all the tasks. This is followed by task-specific heads (orange and green sections), each fine-tuned to capture the unique characteristics of individual tasks, such as Task A and Task B. After fine-tuning, the model is evaluated separately for every task, with iterative adjustment made to optimize its effectiveness across the entire task spectrum.

4. State-of-the-art applications of language models in the cybersecurity domain

The application of language models in cybersecurity has attracted considerable attention in recent research. Broadly speaking, this research can be divided into two main categories: evaluating the capabilities of generic language models in solving cybersecurity problems, and developing customized language models for cybersecurity solutions to provide more efficient and user-friendly tools for security professionals.

4.1. Evaluation of capabilities of generic language models in solving cybersecurity problems

The evaluation of language models for solving cybersecurity problems represents an important milestone in the ongoing development of this field. These models have demonstrated remarkable potential for solving different cybersecurity problems. As indicated by Table 1, researchers are actively exploring language model's ability to understand various cybersecurity concepts (Tann, Liu, Sim, Seah & Chang, 2023) and threat behaviors (Garza et al., 2023), support digital forensics investigation (Scanlon et al., 2023), vulnerability identification and patching (Bakhshandeh, Keramatfar, Norouzi & Mahdi Chekidehkhoun, 2023; Tol & Sunar, 2023), penetration testing (Deng et al., 2023), network security (Jüttner, Grimmer & Buchmann, 2023; Ott, Bogatinovski, Acker, Nedelkoski & Kao, 2021), and generative honeypot (Sladić, Valeros, Catania & Garcia, 2023), and improve overall cybersecurity resilience (Bajaj & Edwards, 2023). The dataset column in the table identifies datasets that were used for prompt engineering (Meng et al., 2023) to assess language model's cybersecurity capabilities.

To explore the potential of language model in understanding the cybersecurity concepts, (Tann et al., 2023) have investigated whether language models can successfully pass a series of professional certification exams in cybersecurity industry and perform well in solving Capture-The-Flag (CTF) exercises. For understanding threat behavior, a study from MIT CSAIL explored the knowledge of large language model (LLM), specifically GPT-3.5, regarding threat behavior in MITER ATT&CK, emphasizing their potential in generating and answering questions about threat behaviors and mitigations (Garza et al., 2023). The study highlights the importance of prompt engineering with context using MITER ATT&CK dataset to provide best answer accuracy with GPT-3.5.

(Scanlon et al., 2023) evaluated the applicability of ChatGPT in the field of digital forensic investigation. They emphasized that due to the phenomenon of 'hallucination' and the sensitivity of the field of digital forensics, caution and thorough examination of the information generated by the ChatGPT model are required.

In 2023, (Bakhshandeh et al., 2023) tested the ability of ChatGPT to identify vulnerabilities in the python source code based on the samples of python code from securityEval and PyT project datasets and

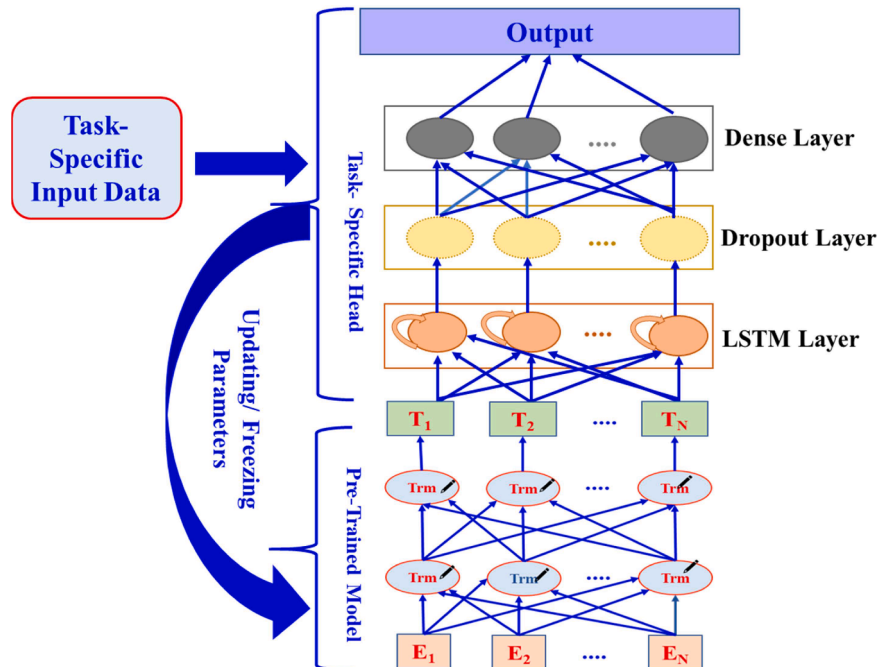


Fig. 5. Transfer learning with task-specific layer.

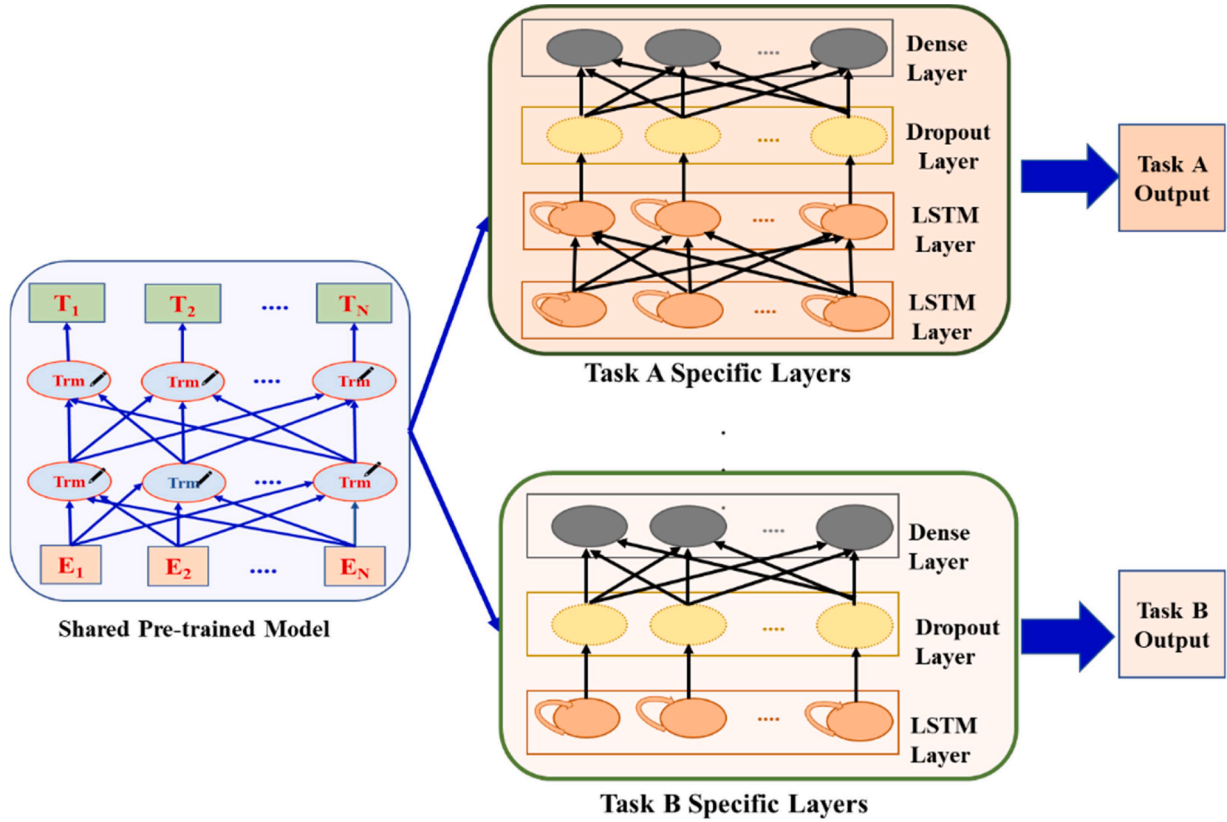


Fig. 6. Multi-task learning.

concluded that the ChatGPT reduces the false positive and false negative rates as compared to the traditional static application security testing tools such as Bandit, Sengrep, and SonarQube. Furthermore, (Tol & Sunar, 2023) explored the capabilities of language models with carefully crafted prompt in generating patches for vulnerable code with microarchitectural side-channel leakage. Additionally, another study by Pearce, Tan, Ahmad, Karri and Dolan-Gavitt (2023) examined the potential of large language models such as Codex, Jurassic-1 and GPT2-csrc, for zero-shot vulnerability repair, demonstrating that these models can generate security patches without prior exposure to specific vulnerabilities.

For penetration testing, researchers (Deng et al., 2023) have introduced a specialized tool to simulate human behavior for automated penetration testing. This tool consists of three self-interacting modules including parsing, reasoning and generation to reserve the context of each conversation to recommend subsequent steps for penetration testing process. Additionally, (Happe & Cito, 2023) demonstrated the use of GPT-3.5 in penetration testing by employing the model for high-level task planning and low-level vulnerability hunting within a vulnerable virtual machine, showcasing the potential of LLMs to augment security testers.

For network security, (Jüttner et al., 2023) proposed ChatIDS that sends anonymized alerts to ChatGPT to explain IDS alerts intuitively and to increase security in the absence of experts. In contrast, (Ott et al., 2021) compared the performance of different language models for obtaining numerical log representation to preserve the semantics of log messages and map them into log vector embeddings suitable for anomaly detection.

Furthermore, researchers (Sladić et al., 2023) are using GPT-3.5 for developing dynamic and realistic software honeypots that convincingly mimic genuine systems. These honeypots can overcome limitations of static honeypots, such as deterministic responses, and lack of adaptability.

To improve active defense in the fight against cybercrime, (Bajaj & Edwards, 2023) evaluated the potential of ChatGPT in the area of scam baiting and concluded that the model must overcome the challenges of following instructions and not provide adversaries with harmful information.

4.2. Development of customized language models for cybersecurity solutions

The development of customized language models for cybersecurity solutions is reshaping the way organizations defend against cyber threats. Unlike traditional AI tools that follow set patterns, language models are trained on extensive textual data to provide a deeper, more context-aware analysis that detects nuanced patterns and anomalies and makes more informed decisions. In addition, customized language models are fine-tuned for specific organizational cybersecurity needs to ensure a tailored defense strategy.

A key benefit is their ability to produce human-readable outputs, making analysis and response more intuitive and actionable for security teams. In the broader context, existing research studies in the area of customized language models for the cybersecurity domain can be divided into four main areas: content analysis, software and system analysis, threat intelligence and monitoring, and cyber vetting, as shown in Table 2. These studies demonstrate how language models combine advanced defense capabilities with clear communication to elevate cybersecurity standards.

To further support this analysis, Table 3 presents the dataset used for fine-tuning and performance metrics used to evaluate the effectiveness of language models in the reviewed studies. These metrics are used widely in cybersecurity research to assess classification performance, anomaly detection, and decision thresholds. These metrics include accuracy, precision, recall, and F1-score, which measure classification performance. Additional metrics such as false alarm rate (FAR),

Table 1

Summary of research studies focused on assessment of capabilities of language models in solving cybersecurity problems.

Author	Purpose	Language Model	Dataset	Use Case	Contribution
Tann et al. (2023)	Knowledge Assessment	ChatGPT, Bard, and Bing chat	Database of questions from Cisco Career Certifications 2023, and CTF (Capture-the-flag) challenges from the events	Understanding of cybersecurity concepts	Assessment of potential of different LLMs in solving cybersecurity CTF exercises and professional certification exams
Garza et al. (2023)	Knowledge Assessment	GPT 3.5	Provided context from MITER ATT&CK for prompt engineering	Understanding of threat behavior and mitigations	Investigation of depth and accuracy of knowledge of threat behavior possessed by LLM
Scanlon et al. (2023)	Knowledge Assessment	ChatGPT	Chat sentiment, hate speech and offensive language dataset for testing automated sentiment analysis	Digital forensic and incident response	Comprehensive examination of the applicability of ChatGPT in digital forensics tasks ranging from simple keyword searching to more complex scenario synthesis
Bakhshandeh et al. (2023)	Capability Assessment	ChatGPT	156 vulnerable python codes from securityEval dataset and PyT project dataset to be used as an input for vulnerability detection using carefully crafted prompts	Vulnerability detection in python source code	Critical assessment of ChatGPT's proficiency in identifying the vulnerabilities within python source code
Tol and Sunar (2023)	Knowledge Assessment	GPT-4, PaLM 2, LLaMA	Docker images of Microwalk packages with version3.1.1-pin for C, and version 3.1.1-jalangi2 for Java script code generation for input	Software vulnerability detection and patching	Exploratory analysis of LLMs to automatically detect and patch side-channel vulnerabilities in software with carefully crafted prompt.
Pearce et al. (2023)	Knowledge Assessment	Codex, Jurassic-1, GPT2-csrc	Customized dataset of code gathered from 10,000 most popular Debian packages.	Vulnerability repair of C/C++ code	Provide insights and prompt design for using LLMs for zero-shot vulnerability repair
Deng et al. (2023)	Knowledge Assessment	GPT-3.5, GPT-4, and Bard	HackTheBox(HTB) and VulnHub for task selection	Penetration testing	Provides the empirical evaluation of selected LLMs for penetration testing and proposed a novel interactive system to provide context
Happe and Cito (2023)	Knowledge Assessment	GPT-3.5.	Used real-world penetration testing scenarios	Penetration testing	Demonstrate the potential of LLM to augment penetration testers in planning and executing penetration tests
Ott et al. (2021)	Capability Assessment	BERT, GPT-2.0, XL	CloudLab OpenStack log dataset	Network security	Leveraged the power of pre-trained language models to transform log messages into a format suitable for deep learning-based anomaly detection.
Jüttner et al. (2023)	Knowledge Assessment	ChatGPT (GPT-3.5 turbo)	Alerts from preconfigured IDS for home network for ChatGPT prompt	Network security	Proposed ChatIDS to explain IDS alerts to non-experts by using ChatGPT to help in understanding potential risks and suggesting countermeasures.
Sladić et al. (2023)	Capability Assessment	GPT-3.5-turbo-16k	An expert for explaining the behavior of real Linux terminal for prompt engineering	Generative honeypots	Utilized LLMs to create credible and dynamic honeypots to overcome the limitations of deterministic responses, and lack of adaptability.
Bajaj and Edwards (2023)	Knowledge Assessment	ChatGPT	Human scam-baiting dataset published by Chen, Wang and Edwards (2023) to provide context for prompt engineering	Email scam-baiting	Evaluation of efficiency of ChatGPT in engaging with and potentially thwarting online scammers

detection rate (DR.), Matthews correlation coefficient (MCC), and AUC evaluate specific aspects of the model performance, including false alarm rates and decision thresholds. Other metrics such as F-micro, F-macro, F-macro std, Loss, and harvest rate provide insights into the specialized tasks or multi-class classification. To ensure clarity, the following formulas detail how these metrics are calculated and interpreted in the reviewed studies:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$False Alarm Rate (FAR) = \frac{FP}{FP + TN}$$

$$Detection Rate (DR) = \frac{TP}{TP + FN}$$

$$F1 - score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

$$True Positive Rate (TPR) = \frac{TP}{TP + FN}$$

$$True Negative Rate (TNR) = \frac{TN}{FP + TN}$$

$$MCC = \frac{(TP * TN) - (FP * FN)}{\sqrt{(TP + FP) * (TP + FN) * (TN + FP) * (TN + FN)}}$$

$$Specificity = \frac{TN}{FP + TN}$$

$$AUC = 1 - Specificity$$

Where, TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, FN is the number of false negatives.

$$F - micro = \frac{2 * Precision_{micro} * Recall_{micro}}{Precision_{micro} + Recall_{micro}}$$

$$\text{Where, } Precision_{micro} = \frac{\sum TP_i}{(\sum TP_i + \sum FP_i)} \text{ and } Recall_{micro} = \frac{\sum TP_i}{(\sum TP_i + \sum FN_i)}$$

$$F - macro = \frac{1}{N} \sum_{i=1}^N F1 - score_i$$

Table 2

Summary of research studies focused on customized language models for cybersecurity solutions.

Author	Purpose	Base Language Model	Fine-Tuning		Use Case	Contribution
			Method	Technique		
Elsadig et al. (2022)	Content Analysis	BERT	Data Centric	Domain Adaptive	Feature extraction	Proposed a novel phishing detection tool that harnesses the power of BERT for feature extraction combined with convolutional neural network (CNN) algorithm for classification
Lee et al. (2021)	Content Analysis	DistilBERT	Parameter Adjustment	Adapter based	Social engineering email detection	Proposed a downsized BERT model named CatBERT for social engineering email detection using content (email body) and context (email header) data
Tida and Hsu (2022)	Content Analysis	BERT	Architecture Extension	Transfer learning with task specific layer	Spam detection	Proposed a universal spam detection model using a variety of spam classification datasets
Xu et al. (2021)	Software and System Analysis	BERT	Architecture Extension	Transfer learning with task specific layer	Malware detection	Proposed a fine-tuned LLM-based method to detect malicious windows software using dynamic analysis.
Demrici et al. (2022)	Software and System Analysis	GPT-2	Architecture Extension	Transfer learning with task specific layer	Malware detection	Proposed LLM-based static analysis technique for malicious software detection
Rahali and Akhloufi (2023)	Software and System Analysis	BERT	Architecture Extension	Transfer learning with task specific layer	Malware detection	Provides an extension of the work proposed by Xu et al. (2021) by providing a tokenizer with feature analyzer on top of the transformer encoder
Souani et al. (2022)	Software and System Analysis	BERT	Parameter Adjustment	Full Model	Malware detection and classification	Replicated the results of technique proposed by Xu et al. (2021) on a more extensive android malware dataset
Thapa et al. (2022)	Software and System Analysis	BERT, MegatronBERT, DistilBERT, RoBERTa, CodeBERT, GPT-2, MegatronGPT-2, GPT-J	Architecture Extension	Transfer learning with task specific layer	Software vulnerability detection	Evaluated the capabilities of transformer-based language models for software vulnerability detection along with the challenges and methodologies of fine-tuning these models for the task.
Omar and Shiaeles (2023)	Software and System Analysis	GPT-2, CodeBERT	Parameter adjustment	Full Model	Software vulnerability detection	Proposed a novel technique for detecting software vulnerabilities using fine-tuned language model
Das et al. (2021)	Software and System Analysis	BERT	Architecture Extension	Transfer learning with task specific layer	Mapping of vulnerabilities to their respective weaknesses	Proposed a framework to improve the mapping of vulnerabilities to hierarchically structured weaknesses, when minimal prior information exists.
Fayyazi and Yang (2023)	Threat Intelligence and Monitoring	GPT 3.5 and SecureBERT with supervised fine-tuning (SFT)	Architecture extension based	Transfer learning with task specific head	Interpretation of ambiguous cyberattack descriptions	Provided a comparison of GPT-3.5 and supervised fine-tuned small-scale LLMs to interpret and forecast description of cyberattacks along with carefully designed prompts
Yin et al. (2020)	Threat Intelligence and Monitoring	BERT	Hybrid (Data Centric + Architecture Extension)	Domain adaptive and Transfer learning with task specific layer	Vulnerability exploitability prediction	Proposed a framework named ExBERT to predict the exploitability of a vulnerability to improve the vulnerability assessment
Setianto et al. (2021)	Threat Intelligence and Monitoring	GPT-2	Parameter Adjustment	Full Model	Log analysis	Presented a run-time tool that leverages a fine-tuned GPT-2 model for parsing dynamic log-lines from a Cowire SSH honeypot.
Manocchio et al. (2023)	Threat Intelligence and Monitoring	BERT and GPT 2.0	Architecture Extension	Transfer learning with task specific layer	Intrusion detection	Proposed a FlowTransformer framework with the flexibility to select the most appropriate hyperparameters and architecture configurations that align with the requirements of network intrusion detection systems.
Seyyar et al. (2022)	Threat Intelligence and Monitoring	BERT	Architecture Extension	Transfer learning with task specific layer	Feature Extraction for Web attack detection	Proposed a novel attack detection framework utilizing BERT for feature extraction and deep learning to discern between normal and anomalous HTTP requests
Aghaei et al. (2022)	Threat Intelligence and Monitoring	RoBERTa	Data Centric	Domain Adaptive	Cyber threat intelligence	Introduced SecureBERT, a versatile, pre-trained language model tailored for diverse cybersecurity tasks, such as enhancing comprehension of cybersecurity reports.
Bayer et al. (2022)	Threat Intelligence and Monitoring	BERT	Data Centric	Domain Adaptive	Cyber threat intelligence	Proposed CYSECBERT, a BERT-based word embedding model for analyzing cybersecurity text, enabling NLP for security domain.
Jin et al. (2023)	Threat Intelligence	RoBERTa	Data Centric	Domain Adaptive	Cyber threat intelligence	Utilized BERT to interpret the specialized language of the Dark Web,

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Table 2 (continued)

Author	Purpose	Base Language Model	Fine-Tuning		Use Case	Contribution
			Method	Technique		
	and Monitoring					facilitating detection of ransomware leaks, highlighting hacking forum threads and extracting threat-related keywords.
Kuehn et al. (2023)	Threat Intelligence and Monitoring	BERT	Data Centric	Domain Adaptive	Automated crawling, classification and ranking of CTI (Cyber threat intelligence) information	Proposed a BERT-based dynamic focused crawler to find and rank CTI-relevant information.
Chen et al. (2021)	Threat Intelligence and Monitoring	BERT	Architecture Extension	Transfer learning with task specific layer	Named entity recognition	BERT-based model for cybersecurity named entity recognition for a combination of Chinese and English text
Zhou et al. (2021)	Threat Intelligence and Monitoring	BERT	Architecture Extension	Transfer learning with task specific layer	Named entity recognition	Proposed a BERT-based method BERT with whole word masking B_{wvm} for named entity recognition (NER) tasks in cybersecurity domain.
Tikhomirov et al. (2020)	Threat Intelligence and Monitoring	BERT	Architecture Extension	Transfer learning with task specific layer	Named entity recognition	BERT-based model for cybersecurity NER in Russian
Evangelatos et al. (2021)	Threat Intelligence and Monitoring	BERT base, ELECTRA base, RoBERTa base, XLNet base	Architecture Extension	Transfer learning with task specific layer	Named entity recognition	Investigated the effectiveness of different LLMs for the extraction of NER in the CTI context
Ranade et al. (2021)	Threat Intelligence and Monitoring	BERT-base-cased	Parameter Adjustment	Full Model	Named entity recognition and multi-class classification	Development of a domain-specific BERT model fine-tuned on the cybersecurity text corpus to perform different cybersecurity-specific tasks
Ameri et al. (2021)	Cyber Vetting	BERT-base	Parameter Adjustment	Full Model	Verification and validation of operational technology infrastructure cybersecurity claims	Proposed a solution to verify and validate the vendor claims about the cybersecurity features of ICS devices to ensure compliance and security in critical infrastructures.

$$F - macro\ std = \sqrt{\frac{1}{N} \sum_{i=1}^N (F1 - score_i - F - macro)^2}$$

Where, N is the number of classes, $F1 - score_i$ is the F1 score of class i

$$Loss_{Cross-entropy} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Where, N is the number of samples, y_i is the actual label for the i^{th} sample (either 0 or 1), p_i is the predicted probability of the positive class for the i^{th} sample

$$Harvest\ rate = \frac{Number\ of\ Relevant\ Documents}{Total\ Number\ of\ Downloaded\ Documents}$$

4.2.1. Content analysis

In cybersecurity, content analysis is essential for the detection of spam and phishing emails. This approach analyzes the header and textual content of the email to find any anomalies or patterns that indicate malicious intent.

For customized language models, researchers have fine-tuned the pre-trained models using datasets with different legitimate and fraudulent emails to provide a refined approach to differentiate between legitimate content and fraudulent spam or phishing attempts (Elsadig et al., 2022; Lee, Saxe, Harang & Al, 2021; Tida & Hsu, 2022). In this direction, Lee et al. (2021) proposed CatBERT by downsizing an existing LLM (DistilBERT), and using adapter-based fine-tuning for phishing detection. In contrast, Elsadig et al. (2022) used a BERT model for feature extraction from URL data for phishing site prediction. Furthermore, in 2022, Tida and Hsu (2022) developed a universal model for spam detection, which has used transfer learning with task-specific layers for fine-tuning the pre-trained BERT base model. They have used a classification head consisting of linear layers, dropout layers,

batch normalization layers, rectified linear unit activations, a logarithmic SoftMax activation, and a Xavier initialization for weights.

4.2.2. Software and system analysis

Software and system analysis is the process of scrutinizing software applications and system configurations to uncover malware, potential penetration points, and vulnerabilities. With the integration of customized language models, this scrutiny is elevated due to their ability to process vast amounts of data at unprecedented speeds, combined with the deep understanding of patterns and anomalies. The fine-tuning can further empower them to discern intricate malware behavior and code analysis (Demrici et al., 2022; Rahali & Akhloufi, 2023; Souani, Khanfir, Bartel, Allix & Le Traon, 2022; Xu, Fang & Yang, 2021), and highlight vulnerabilities (Das, Serra, Halappanavar, Pothan & Al-Shaer, 2021; Omar & Shiaeles, 2023; Thapa et al., 2022) that might be overlooked by conventional tools.

Till date, a number of researchers have introduced the use of customized BERT and GPT 2.0 for malware detection using static, dynamic and hybrid analysis. In static analysis, where code is examined without execution, language models were trained to recognize patterns or signatures indicative of malware within vast codebases (Demrici et al., 2022; Rahali & Akhloufi, 2023; Souani et al., 2022). Demrici et al. (2022) have used a linear layer as a language modeling head along with careful hyperparameter tuning for fine-tuning the GPT-2 model for malware detection. Similarly, (Rahali & Akhloufi, 2023) proposed MalBERTv2 by utilizing the pre-trained BERT model's extensive knowledge and extend it with additional layers specific to the malware classification task along with careful parameter optimization. In contrast, Souani et al., 2022 tested the performance of BERT and MalBERT for android malware detection by utilizing full-model fine-tuning technique for parameter adjustment with android malware datasets. On the other hand, in dynamic analysis, software behavior was observed during runtime to analyze system logs, interactions and behaviors to

Table 3

Performance metrics used in research studies focused on customized language models for cybersecurity solutions.

Author	Dataset Used for Fine-tuning	Performance Metrics Used	Performance Reported																											
Elsadig et al. (2022)	Phishing site predict dataset by Kaggle	Accuracy, F1-score, Precision	Accuracy = 96.66%, F1-score = 93.63%, Precision = 96.66%																											
Lee et al. (2021)	Industrial-scale datasets from Sophos cloud email security service	AUC	0.9894																											
Tida and Hsu (2022)	Enron, Spamassain, Lingspam, and Spamtext message classification datasets	Accuracy, F1-score, Precision, Recall	Accuracy = 0.97, F1-score = 0.96, Precision = 0.95, Recall = 0.97																											
Xu et al. (2021)	Ki dataset and Catak dataset	Accuracy, Precision, Recall, F1-score, TPR, TNR	Accuracy = 99.98%, Precision = 100%, Recall= 99.98%, F1-score = 99.99%, TPR = 99.98%, TNR = 100%																											
Demrici et al. (2022)	Datasets of portable executable (PE) files, including Win32, Commando VM v-2.0, and sore1–20 m, along with VirusShare website data.	Accuracy	85.4%																											
Rahali and Akhloufi (2023)	MixG-Androzoo, MixG-VirusShare, MixG-AMD, and MixG-Derbin	Accuracy, Precision, Recall, F1-score, MCC, AUC	Accuracy = 97.1%, Precision= 98.4%, Recall= 96.7%, F1-score = 97.2%, MCC = 95.8% AUC= 98.6%																											
Souani et al. (2022)	AndroZoo dataset	Accuracy, Precision, Recall, F1-score, Loss	Accuracy = 0.970, Precision = 0.957, Recall= 0.941, F1-score = 0.949, Loss= 0.183																											
Thapa et al. (2022)	VulDeePecker Dataset	Precision, Recall, F1-score	The results indicate that GPT-2 Large and GPT-2 XL have best F1-score for both binary and multi-class classification.																											
			<table><tr><th rowspan="2">Model</th><th colspan="3">Binary</th><th colspan="3">Multi</th></tr><tr><th>P</th><th>Re</th><th>F1</th><th>P</th><th>Re</th><th>F1</th></tr><tr><td>GPT-2 L</td><td>96.3</td><td>96.1</td><td>96.2</td><td>96.9</td><td>95.9</td><td>95.9</td></tr><tr><td>GPT-2 XL</td><td>96.2</td><td>95.9</td><td>96</td><td>96.2</td><td>95.9</td><td>96.1</td></tr></table>	Model	Binary			Multi			P	Re	F1	P	Re	F1	GPT-2 L	96.3	96.1	96.2	96.9	95.9	95.9	GPT-2 XL	96.2	95.9	96	96.2	95.9	96.1
Model	Binary				Multi																									
	P	Re	F1	P	Re	F1																								
GPT-2 L	96.3	96.1	96.2	96.9	95.9	95.9																								
GPT-2 XL	96.2	95.9	96	96.2	95.9	96.1																								
Omar and Shiaeles (2023)	SARD (Software Assurance Reference Dataset) and SeVC dataset	Accuracy, F1-score, AUC	<table><tr><th rowspan="2">Model</th><th colspan="3">SARD</th><th colspan="3">SeVC</th></tr><tr><th>Ac</th><th>F1</th><th>AUC</th><th>Ac</th><th>F1</th><th>AUC</th></tr><tr><td>GPT-2</td><td>92.6</td><td>92.4</td><td>90.3</td><td>87.6</td><td>86.9</td><td>91.5</td></tr><tr><td>CodeBERT</td><td>89.3</td><td>89.9</td><td>94.1</td><td>83.2</td><td>76.5</td><td>79.8</td></tr></table>	Model	SARD			SeVC			Ac	F1	AUC	Ac	F1	AUC	GPT-2	92.6	92.4	90.3	87.6	86.9	91.5	CodeBERT	89.3	89.9	94.1	83.2	76.5	79.8
Model	SARD				SeVC																									
	Ac	F1	AUC	Ac	F1	AUC																								
GPT-2	92.6	92.4	90.3	87.6	86.9	91.5																								
CodeBERT	89.3	89.9	94.1	83.2	76.5	79.8																								
Das et al. (2021)	Common Vulnerabilities and Exposure (CVE) dataset, collected from the NVD website for the years 1999–2017 (Test1), 2018–2020 (Test 2), and Common weakness exposure dataset from MITER	Accuracy, F1-score	<table><tr><th>Dataset</th><th>Accuracy</th><th>F1</th></tr><tr><td>Test1</td><td>89-98%</td><td>0.93</td></tr><tr><td>Test 2</td><td>81-95%</td><td>0.92</td></tr></table>	Dataset	Accuracy	F1	Test1	89-98%	0.93	Test 2	81-95%	0.92																		
Dataset	Accuracy	F1																												
Test1	89-98%	0.93																												
Test 2	81-95%	0.92																												
Fayyazi and Yang (2023)	MITER ATT& CK and Common attack pattern enumeration and classification (CAPEC) datasets for fine-tuning and prompt engineering	Accuracy, Micro F1-score	<table><tr><th>Model</th><th>Accuracy</th><th>Micro-F1</th></tr><tr><td>SecureBERT-SFT</td><td>0.92</td><td>0.97</td></tr><tr><td>GPT-3.5</td><td>0.44</td><td>0.67</td></tr></table>	Model	Accuracy	Micro-F1	SecureBERT-SFT	0.92	0.97	GPT-3.5	0.44	0.67																		
Model	Accuracy	Micro-F1																												
SecureBERT-SFT	0.92	0.97																												
GPT-3.5	0.44	0.67																												
Yin et al. (2020)	Dataset of NVD and ExploitDB published between 1999 and 2019	Accuracy, Precision, Recall, F1-score	Accuracy = 0.911, Precision = 0.918, Recall= 0.905, F1-score = 0.912																											
Setianto et al. (2021)	CyberLab honeynet dataset	F1-score	0.89																											
Manocchio et al. (2023)	NSL-KDD, UNSW-NB15, CSE-CIS-IDS2018 datasets	F1-score, Detection rate (DR.), False alarm rate (FAR)	<table><tr><th colspan="2" rowspan="2">Model</th><th>F1</th><th>DR</th><th>FAR</th></tr><tr><th></th><th></th><th></th></tr><tr><td rowspan="2">GPT</td><td>Comb1</td><td>94.9</td><td>97.9</td><td>0.73</td></tr><tr><td>Comb2</td><td>94.9</td><td>97.7</td><td>0.84</td></tr><tr><td rowspan="2">BERT</td><td>Comb1</td><td>92.7</td><td>95.6</td><td>0.77</td></tr><tr><td>Comb2</td><td>88.4</td><td>94.7</td><td>1.55</td></tr></table>	Model		F1	DR	FAR				GPT	Comb1	94.9	97.9	0.73	Comb2	94.9	97.7	0.84	BERT	Comb1	92.7	95.6	0.77	Comb2	88.4	94.7	1.55	
Model		F1	DR			FAR																								
GPT	Comb1	94.9	97.9	0.73																										
	Comb2	94.9	97.7	0.84																										
BERT	Comb1	92.7	95.6	0.77																										
	Comb2	88.4	94.7	1.55																										
Seyyar et al. (2022)	CSIC 2010, FWAf, and HttpParams dataset	F1-score	98.07%																											
Aghaei et al. (2022)	Dataset contains various forms of cybersecurity texts, ranging from basic information, news to more advanced texts such as research articles, and threat reports.	Error, Accuracy, F1-score, Precision, Recall	<table><tr><th rowspan="2">Model</th><th colspan="3">Sentiment Analysis</th><th colspan="3">NER</th></tr><tr><th>Error</th><th>Acc</th><th>F1</th><th>Pr</th><th>Re</th><th>F1</th></tr><tr><td>SecureBERT</td><td>0.77</td><td>67.2</td><td>67.1</td><td>85.2</td><td>88.1</td><td>86.6</td></tr></table>	Model	Sentiment Analysis			NER			Error	Acc	F1	Pr	Re	F1	SecureBERT	0.77	67.2	67.1	85.2	88.1	86.6							
Model	Sentiment Analysis				NER																									
	Error	Acc	F1	Pr	Re	F1																								
SecureBERT	0.77	67.2	67.1	85.2	88.1	86.6																								

(continued on next page)

Table 3 (continued)

Author	Dataset Used for Fine-tuning	Performance Metrics Used	Performance Reported																																												
Bayer et al. (2022)	Crawled dataset from twitter, blog posts, arXiv articles, NVD, MS Exchange, and CySecAlert datasets	F1-score	<table><tr><th rowspan="3">Model</th><th colspan="5">F1-Score</th></tr><tr><th colspan="3">NER</th><th colspan="2">Classification</th></tr><tr><th>SV</th><th>SN</th><th>AC</th><th>MS Exchange</th><th>CySecAlert</th></tr><tr><td>CySecBERT</td><td>0.93</td><td>0.89</td><td>0.35</td><td>0.89</td><td>0.89</td></tr></table>	Model	F1-Score					NER			Classification		SV	SN	AC	MS Exchange	CySecAlert	CySecBERT	0.93	0.89	0.35	0.89	0.89																						
Model	F1-Score																																														
	NER				Classification																																										
	SV	SN	AC	MS Exchange	CySecAlert																																										
CySecBERT	0.93	0.89	0.35	0.89	0.89																																										
Jin et al. (2023)	Used seed addresses from Ahmia and public repositories containing lists of onion domains, and dark web text datasets known as DUTA and CoDA	Precision, Recall, F1-Score	Here, SV, SN, and AC are software versions, software names, and attack complexities of NVD descriptions, respectively. <table><tr><th colspan="2">Use Case</th><th>Pr</th><th>Re</th><th>F1</th></tr><tr><td rowspan="2">Ransomware Leak Site Detection</td><td>Raw (R)</td><td>78.8</td><td>83.6</td><td>79.9</td></tr><tr><td>Pre-processed (P)</td><td>85.2</td><td>84.6</td><td>84.1</td></tr><tr><td rowspan="4">Dark web Activity classification</td><td rowspan="2">DUTA</td><td>R</td><td>80.1</td><td>79.9</td><td>80</td></tr><tr><td>P</td><td>79.9</td><td>80.1</td><td>79.9</td></tr><tr><td rowspan="2">CoDA</td><td>R</td><td>94.1</td><td>94.4</td><td>94.3</td></tr><tr><td>P</td><td>94.3</td><td>94.3</td><td>94.3</td></tr><tr><td rowspan="2">Thread Detection</td><td colspan="2">R</td><td>75.9</td><td>43.1</td><td>52.9</td></tr><tr><td colspan="2">P</td><td>72.4</td><td>45.1</td><td>54.2</td></tr></table>	Use Case		Pr	Re	F1	Ransomware Leak Site Detection	Raw (R)	78.8	83.6	79.9	Pre-processed (P)	85.2	84.6	84.1	Dark web Activity classification	DUTA	R	80.1	79.9	80	P	79.9	80.1	79.9	CoDA	R	94.1	94.4	94.3	P	94.3	94.3	94.3	Thread Detection	R		75.9	43.1	52.9	P		72.4	45.1	54.2
Use Case		Pr	Re	F1																																											
Ransomware Leak Site Detection	Raw (R)	78.8	83.6	79.9																																											
	Pre-processed (P)	85.2	84.6	84.1																																											
Dark web Activity classification	DUTA	R	80.1	79.9	80																																										
		P	79.9	80.1	79.9																																										
	CoDA	R	94.1	94.4	94.3																																										
		P	94.3	94.3	94.3																																										
Thread Detection	R		75.9	43.1	52.9																																										
	P		72.4	45.1	54.2																																										
Kuehn et al. (2023)	Customized dataset with information about TTPs, malware used, broad information, and vulnerability targeted.	Harvest rate	51.2%																																												
Chen et al. (2021)	Customized dataset from the Cyber Security Laboratory of the School of Computer Science and Technology, Guizhou University	Accuracy, Precision, Recall, F1-Score	Overall, BERT-Bi-LSTM-CRF performs best with Accuracy = 98.23, Precision= 94.77, Recall= 92.97, and F1-score = 93.11																																												
Zhou et al. (2021)	Dataset provided by Bridges, Jones, Iannacone, Testa and Goodall (2013) which covers the NVD, Microsoft security bulletin, and Metasploit framework	Precision, Recall, F1-Score	Precision = 97.03, Recall = 96.71, and F1-score = 96.87																																												
Tikhomirov et al. (2020)	Sec_col corpus (Sirotna and Loukachevitch (2019))	F-micro, F-macro, F-macro std	F-micro = 72.74, F-macro = 68.82, F-macro std = 0.86																																												
Evangelatos et al. (2021)	Dataset for NER in Threat Intelligence	F1-macro, Precision _{macro} , Recall _{macro}	Concluded that XLNet-base model performs best with f1-macro = 0.883, Precision _{macro} = 0.863, and Recall _{macro} = 0.906																																												
Ranade et al. (2021)	Customised cybersecurity text corpus from krebs on security, APT reports, CVE database, and NVD	F1-score, Precision, Recall, and Loss	F1 = 0.811 Precision = 0.802 Recall = 0.823 Loss = 0.531																																												
Ameri et al. (2021)	Custom dataset containing claim related sequences curated from ICS device documentation	Accuracy, F1-macro, AUC	Accuracy = 0.954, F1-score =0.93, AUC = 0.948																																												

detect malicious activities. In this direction, Xu et al. (2021) have adapted the pre-trained BERT model for malware classification task by integrating a classification module into its output layer. They then trained this modified model with labelled data to improve its classification capabilities.

For vulnerability assessment, Thapa et al. (2022) provided an empirical analysis of language models for software vulnerability detection along with the analysis of popular platforms to efficiently fine-tune these models. They have used different classification head comprising of different number of linear and dropout layers on the top of different pre-trained models. Also, Das et al. (2021) have proposed a BERT-based framework to efficiently map common vulnerabilities and exposures (CVEs) reports to hierarchically structured common weakness enumerations (CWEs) to efficiently classify diverse vulnerabilities using custom layers with dropout and fully connected neural networks on the top of the pooling layer to predict all usable CWEs. Furthermore, Omar and Shiaeles (2023) introduced VulDetect, a novel transformer-based vulnerability detection framework by fine-tuning GPT-2 on different

benchmarked datasets of vulnerable code.

4.2.3. Threat intelligence and monitoring

Threat intelligence and monitoring are critical components of cybersecurity that focus on the collection, analysis, and dissemination of information about emerging threats to offer a clear picture of lurking dangers in the cyber landscape. For real-time monitoring, customized language models can sift through logs, detect patterns and anomalies rapidly. Moreover, their advanced natural language processing capabilities enable them to interpret unstructured data from sources like social media and online forums, where threat actors might discuss vulnerabilities or share malicious strategies (Kereopa-Yorke, 2023). To improve threat detection capabilities, researchers are thus actively investigating customized language models, particularly in areas such as intrusion detection (Manocchio et al., 2023), web attack detection (Seyyar, Yavuz & Ünver, 2022), log analysis (Setianto et al., 2021), generative honeypots (Sladić et al., 2023), and cyber threat intelligence (Aghaei, Niu, Shadid & Al-Shaer, 2022; Bayer et al., 2022; Chen, Ding, Li

& Chen, 2021; Jin et al., 2023; Zhou et al., 2021).

Because of their adaptability, language models can be used to design customized intrusion detection systems for specific network environments and traffic patterns with the generation of human understandable and actionable alerts. For this, Manocchio et al., 2023 proposed a FlowTransformer framework to implement network-based intrusion detection system by integrating a custom classification head, including advanced layers and special processing units for feature wise projection. Additionally, a unique classification token is integrated in the model for more precise and context-aware classification outcomes.

Customized language models can transform raw log data into actionable intelligence by presenting it in a human readable format. In 2021, Setianto et al., 2021 fine-tuned the GPT-2 model on the existing corpus of illegal Unix commands to analyze the dynamic logs generated by Cowire SSH honeypot instance for predictive analytics. The main focus is on tweaking the training parameters (like epochs, batch size, and memory considerations) and selecting a model variant that is best suited to the data and task at hand, without significant changes to the model's architecture.

By integrating language models, organizations can greatly enhance their web attack detection systems and distinguish between benign and malicious HTTP requests. For this, Seyyar et al., 2022 have used a BERT tokenizer to extract features from text input, and MLP layers are further used to process and classify the features derived from BERT for better web attack detection.

Language models also provide an innovative approach to understanding and anticipating the ever-changing threat landscape by extracting subtle insights, and emerging threats from various threat intelligence sources to predict potential attack vectors. For this, researchers (Aghaei et al., 2022; Bayer et al., 2022; Jin et al., 2023) have introduced specialized language models for solving crucial NLP tasks in cybersecurity such as information extraction from threat reports, dark web, social sites and discussion forums. They have mainly focused on the domain adaptive fine-tuning technique of data centric method for retraining a general language model with specialized domain-specific data to enhance its relevance and effectiveness in cybersecurity domain. In this direction, researchers (Chen et al., 2021; Evangelatos et al., 2021; Tikhomirov, Loukachevitch, Sirotina & Dobrov, 2020; Zhou et al., 2021) have used customized BERT-based models for cybersecurity named entity recognition (NER) to identify specific entities in text data, such as IP addresses, domain names, malware hashes, or threat actor names. The fine-tuning methods described indicated the use of architecture extension-based fine-tuning, where the existing BERT based model is augmented with additional task specific layers to adapt it for specific NER task. In contrast, Kuehn, Schmidt and Reuter (2023) proposed a framework to find and rank CTI-relevant information by downloading and classifying documents. This document classification is achieved by using domain adaptive fine-tuning to tailor the language model to the specific cybersecurity tasks. Additionally, Ranade, Piplai, Joshi and Finin (2021) introduced CyBERT, a BERT based model fine-tuned with large corpus of cybersecurity textual data, extending BERT's vocabulary and demonstrating its effectiveness in tasks such as cybersecurity NER, multi-class classification, and cybersecurity knowledge graph completion, significantly outperforming the base BERT model in these tasks. Furthermore, Yin, Tang, Cao and Wang (2020) have proposed a BERT-based prediction framework named ExBERT to accurately predict if a vulnerability will be exploited or not. The proposed framework used a data-centric approach to fine-tune the model's understanding of domain-specific content and extends the model's architecture by adding new layers for specialized processing and prediction tasks. Also, Fayyazi and Yang (2023) compare the performance of direct use of language models vs. fine-tuned language models for interpretation of ambiguous descriptions of MITER ATT&CK. They highlight the limitations of direct use of language models in interpreting multi-label, ambiguous tactics, techniques, and procedures (TTP) types of descriptions.

4.2.4. Cyber vetting

Cyber vetting refers to the systematic process of assessing, scrutinizing, and confirming the authenticity and accuracy of cybersecurity measures and practices claimed by operational technology (OT) and information technology systems. In this direction, Ameri, Hempel, Sharif, Lopez and Perumalla (2021) developed a semi-automated vetting engine by fine-tuning the BERT language model for cyber physical security assurance to verify and validate vendor's OT infrastructure cybersecurity claims at both pre-deployment and post-deployment periods. The fine-tuning method centered around parameter adjustment-based fine-tuning for adapting the model's parameter and training process to the nuances of the data and task.

5. Development and dimensions of customized language model for cybersecurity

To empower the cybersecurity domain with advanced language model's capabilities, there is a growing need to bridge the gap between vast data realms and actionable insights. This means providing organizations with a dynamic tool that can adapt, predict, and defend against evolving threats. Customized language models, defined as fine-tuned open language models, are at the forefront of this initiative. To lay the foundation for such an endeavor, it is essential to explore the dimensions of these fine-tuned models along with an outline of development and use of these customized models.

5.1. Dimensions of customized language models for cybersecurity domain

To provide a more structured approach for the development and use of these customized models, specific dimensions like services, data, processes, and users are discussed in detail as highlighted in Fig. 7.

The **data dimension** serves as the core foundation for the customized model to understand the cybersecurity terminologies and nuances, and contextualize the model to generate informed, accurate, and domain-specific responses tailored to the intricate nuances of the cybersecurity landscape. Here, vast amount of cybersecurity data is collected, processed, and analyzed, ranging from threat data to the event logs. The nature and quality of the data determine the reliability, adaptability, and relevance of the framework in the ever-evolving world of cybersecurity. Thus, proper data acquisition, management, and continuous updating are the paramount for the success of the customized language models. This can be ensured by cleaning, integrating and transforming the raw data into contextualized data by using different data-preprocessing techniques.

The **processes dimension** encompasses the methodologies, workflows, and protocols that guide knowledge extraction, model customization, deployment, and maintenance in the realm of cybersecurity. The choice of customization methods depends on the specific task requirements, the availability of domain-specific data, and the desired balance between model performance and resource constraints. In contrast, adaptive learning is a continuous process of adaptation in which the model evolves with new data or user input to maintain its applicability and efficacy for ever-changing cybersecurity threats.

For the **services dimension**, the applicability of the customized model can be seen in various cybersecurity use cases, ranging from content analysis to more complex proactive threat intelligence and monitoring, as explained in Section 4. When selecting the best language model use cases for cybersecurity, it is critical to balance need and risk. Organizations looking to incorporate language models into their cybersecurity must weigh the risk of misinformation and errors against the technology requirements. This will ensure the effectiveness and integrity of customized language models for cybersecurity.

The true potential of the model is seen in the **user dimension**, where the user interface is developed with the end user in mind, regardless of whether they are an experienced cybersecurity analyst or an IT novice.

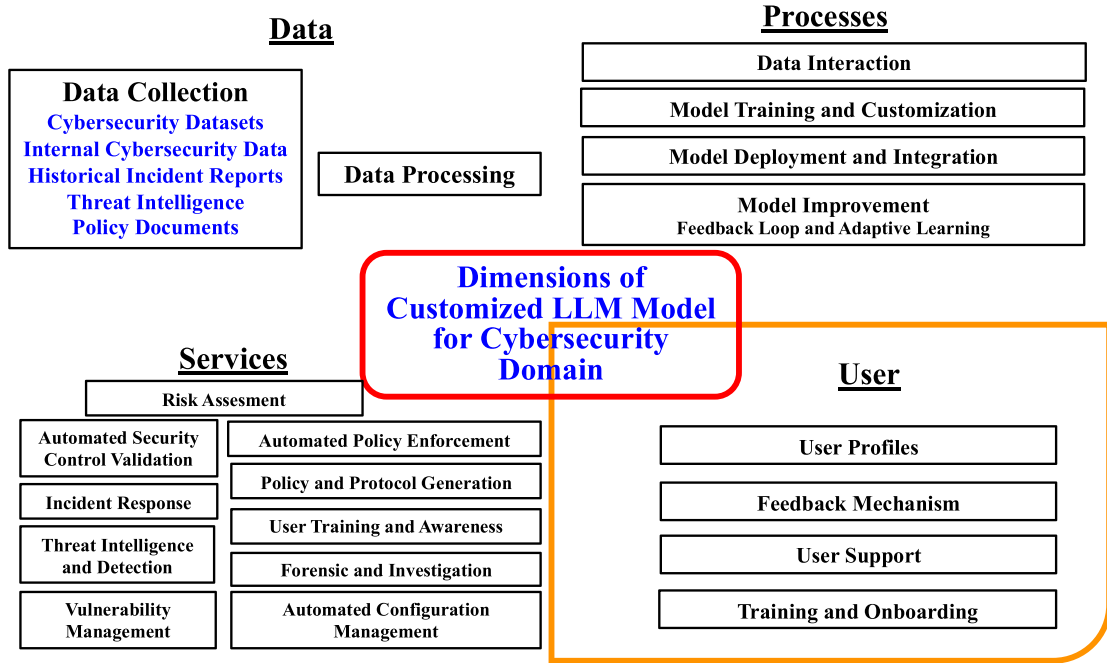


Fig. 7. Distinct dimensions of customized language model for cybersecurity domain.

5.2. Outline for the development of a customized language model for cybersecurity

As shown in Fig. 8, customizing a language model for the cybersecurity is a multi-layered process in which each stage needs to be carefully considered. The first step in the process is a thorough understanding of domain expertise. Thus, it is essential to involve cybersecurity experts who have knowledge of threat landscapes, attack vectors and security tactics. Their insights are extremely useful in developing a language model that is customized according to the unique challenges of the cybersecurity domain.

The next step revolves around the data. It is crucial to curate a high-quality dataset specific to the chosen cybersecurity service. The dataset should contain a wide range of relevant data depending on the cybersecurity context, whether it's attack detection, phishing attempts, or

another scenario. This may include malware samples, historical incident reports, network logs, and any other data relevant to the specific cybersecurity service selected for implementation. The data needs to be pre-processed to deal with missing values, anomalies and data imbalance that are common in cybersecurity datasets. Ethical and regulatory considerations in data collection, as well as accurate and consistent labeling of data, are the biggest obstacles in the evolving field of cybersecurity.

In the model selection and setup stage, the language model is selected based on its ability to process the dataset and the complexity of the task. Every model excels in specific tasks, for example BERT understands context, while GPT generates coherent text. Open-source LLMs, in particular, are gaining traction, enabling developers to create customizable solutions cost-effectively. Selecting the right model for the task is crucial for successful development of a customized language

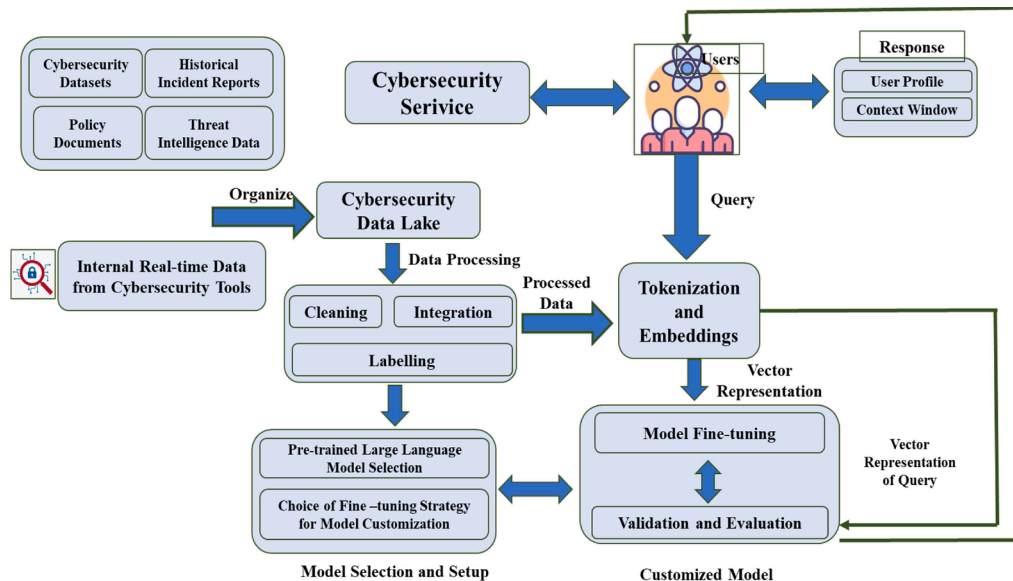


Fig. 8. Outline of customized language model for cybersecurity.

model. Further, the model can be customized by using the fine-tuning strategies explained in Section 3.

The next step is important to convert the data into the format expected by the language model. In this step, the dataset is tokenized to translate the domain's unique text patterns, such as email text, or code snippets into distinct, analyzable tokens, which are processed by the LLM during training and inference. These tokens are then transformed into an initial vector representation through the embedding layer of selected language model, effectively capturing both the semantic relationships and syntactic properties of each token. This process ensures that the model can understand the context and nuances within the input data.

Next, the model is fine-tuned based on the embedded data using the techniques from Section 3. This involves a series of targeted adjustments to improve the performance of the model for specific tasks. Transfer learning concepts and regularization techniques are used as they are essential to prevent overfitting for models trained on highly specific cybersecurity datasets.

Rigorous evaluation is required to verify the effectiveness of the model. This includes using appropriate metrics such as precision, recall, and F1 score, as well as performing cross-validation and testing the model against real data to ensure its robustness and reliability. To ensure the robustness of the model against fraudulent inputs that mimic cybersecurity threats, adversarial testing should be added. The interpretability of the model is also crucial. Cybersecurity professionals need to understand why the model makes certain decisions. Techniques such as feature mapping methods help to achieve this and verify that the model is focusing on the right features of the task.

The process of development and validation of a model is followed by its deployment and integration. For practical applications, the language model must be integrated with existing cybersecurity workflows and tools. An essential component of this stage is the creation of APIs and interfaces to enable language model's interaction with security operations.

The incorporation of user profiles into the customized model can enhance its capacity to offer personalized recommendations aligned with organizational policies and user responsibilities. These user profiles contain information about the user's role in the organization, their access rights, and their prior interactions with security alerts or incidents. The process of integrating user profiles into the customized language models includes identifying user profiles, maintaining these profiles, and modifying the model's interaction based on these profiles. This process also includes modifying the language model response generation mechanism to adjust the technical complexity of the language, filtering information based on role relevance, or highlighting information relevant to the user's tasks.

Furthermore, in order to understand and respond to cybersecurity events effectively, it is essential to define and maintain an adequate context window during user interactions. This enhances user experience and relevance by guaranteeing that the model's responses are appropriately informed by the history of interactions.

The cybersecurity landscape is dynamic and ever-evolving due to the frequent emergence of new threats. Therefore, implementing systems for continuous learning and adaptation is crucial. Regular updates with new threat and incident data ensure that the model continues to effectively detect and defend against new threats that were not included in the original training data set. This is usually ensured by incrementally fine-tuning the custom language model to new data to reflect recent changes in the field.

In conclusion, customizing a language model for cybersecurity is a careful and comprehensive operation. It involves expertise, data maintenance, model selection and setup, fine-tuning, interpretability, rigorous evaluation, deployment and integration, and regular updates. This customization enables organizations to leverage the power of language models to strengthen their cybersecurity defenses and successfully counter the ever-evolving threats.

6. Challenges and considerations

A comprehensive examination of challenges and considerations involved with developing and deploying a customized language model for cybersecurity is necessary to ensure its reliability and effectiveness. These challenges not only underscore potential technological barriers and risks but also highlight critical socio-technical issues that influence organizational processes and decision making. The effective implementation and operation of such models requires consideration of both the technical aspects and the human, organizational, and ethical factors that influence the usability of language models in cybersecurity. By identifying and understanding these challenges, and integrating technical, organizational, and human-centric approaches, the overall reliability and effectiveness of the cybersecurity language models can be improved. Below is the detailed explanation of the main challenges and the corresponding considerations:

- **Data Confidentiality:** The foundation of customized language model for cybersecurity lies in its ability to process and understand large data sets, which frequently include incident reports, sensitive logs, and confidential organizational data (Brown et al., 2022; Kasneci et al., 2023). The inclusion of such data raises concerns about the potential for unauthorised access, data leaks, and misuse (Dong, Moon, Xu, Malik & Yu, 2023; Gupta, Akiri, Aryal, Parker & Praharaj, 2023; Yao et al., 2024). One of the primary risks is data leakage, where sensitive information could be inadvertently revealed through model's output (Gupta, Nair, Mishra, Ibrahim & Bhardwaj, 2024). Attackers might exploit sophisticated queries to extract confidential information from the model, which could lead to severe privacy breach. Importantly data privacy and protection are not only technical challenges, but also ethical and socio-technical issues that require both technological safeguards and human oversight (Dwivedi et al., 2021).

To mitigate these risks, it is essential to implement a range of robust strategies that address both technological and human factors. It includes techniques such as differential privacy, regular audits and continuous monitoring, access control and encryption, and adversarial testing. *Differential privacy* techniques can ensure that the output of the model does not reveal sensitive information about individual data points in the training set, thereby protecting user's privacy even when the model is queried extensively (Yao et al., 2024). *Regular audits and monitoring* of the model's output are crucial to immediately identify and address unintentional disclosures of sensitive information. Additionally, implementing strict *access control and encryption* for both data at rest and in transit can significantly reduce the risk of unauthorised access and data leaks (Pool, Akhlaghpour, Fatehi & Burton-Jones, 2024).

Furthermore, *adversarial testing*, where ethical hackers attempt to exploit the model, can help identify vulnerabilities that need to be addressed (Dwivedi et al., 2021). In combination, these strategies, not only improve the technical security of language models in cybersecurity applications, but also take into account the organizational and ethical aspects of data confidentiality. By incorporating socio-technical considerations, such as the interplay between human oversight and automated processes, these models can be deployed in a way that improves automation and threat detection without introducing new vulnerabilities or compromising data confidentiality (Gupta et al., 2024). This approach ensures that language models become more secure, reliable, and aligned with the broader socio-technical landscape of cybersecurity.

- **Infrastructure Requirement:** The advanced capabilities of customized language models designed for cybersecurity requires a robust computational framework that includes both processing power and storage capacity (Kasneci et al., 2023). As the model ingests and processes massive amounts of cybersecurity data, it requires high-performance servers, possibly equipped with GPU

acceleration, to ensure timely and efficient operations. In addition to pure hardware requirements, specialized software frameworks and tools are also needed to optimize data management, pre-processing and model training. As organizations grow, evolve, and face an ever-expanding threat landscape, the volume and complexity of the data they process can increase dramatically. A major challenge is the scalability of the infrastructure. As data volumes increase, it becomes increasingly difficult to maintain performance levels. Specific risks include system bottlenecks, data processing delays, and reduced real-time threat detection accuracy. In addition, the energy consumption of high-performance computers raises sustainability issues. The use of *energy-efficient hardware* and *cloud-based infrastructure* can be critical to overcoming these challenges (Kasneci et al., 2023). Cloud solutions offer flexible scalability, allowing companies to scale resources up or down as needed, optimizing costs and performance. *Regular infrastructure reviews and upgrades* are also necessary to ensure that the system can meet growing data and computing demands (Saha et al., 2024). This proactive approach can help in the identification of potential bottlenecks and inefficiencies before they impact the operation. By addressing these constraints with strategic investments in infrastructure and technology, organizations can ensure the sustainable operation and scalability of cybersecurity language models, ensuring their longevity and adaptability in an ever-evolving landscape.

- Evolving Threat Landscape:** Cyber attackers are constantly developing new tactics, techniques, and procedures to infiltrate defenses. The threats of today can change greatly from those of tomorrow because of this evolution (Sun et al., 2023). This ever-changing landscape poses significant risks to customized language models in cybersecurity, primarily due to their reliance on historical data that can quickly become outdated. Specific risks include model obsolescence, ineffective threat detection and increased vulnerability to zero-day attacks. Overcoming these challenges requires regular updates and adaptive training, but they also have drawbacks (Lira, Marroquin & To, 2024). *Regular updates* include regular training with the latest threat intelligence and associated data to ensure the model remains relevant. However, this process can be resource intensive and may not keep pace with the rapid evolution of threats, leading to gaps in protection. *Adaptive training*, which enables dynamic adaptation to new threats, is technically demanding and cost-intensive. It requires continuous learning, significant computing power and sophisticated algorithms, which can overwhelm an organization's resources and infrastructure. The complexity of implementing adaptive training can also lead to inconsistent performance and potential system failures if not properly managed. Prioritizing critical updates, investing in a scalable cloud-based infrastructure, leveraging collaborative threat intelligence, deploying hybrid update strategies and conducting regular audits and performance monitoring are important strategies to address these challenges. By implementing these measures, organizations can improve the robustness, agility and readiness of adapted language models to defend against the ever-changing cybersecurity attacks, despite the difficulties and costs involved.
- Integration Complexity:** Organizations use a variety of systems and tools, each serving a specific function, such as intrusion detection or incident response. To successfully implement customized language model into cybersecurity environment, technical expertise and careful planning is required (Mikhailov, 2023). These models must process data from heterogeneous sources and integrate seamlessly with existing security tools (Scanlon et al., 2023). The complexity of integration poses specific risks such as business disruption, data incompatibility, and workflow inefficiencies. If these risks are not properly managed, they can lead to increased vulnerabilities and reduced effectiveness of cybersecurity operations. To mitigate these challenges, organizations should develop a *comprehensive integration plan* with compatibility assessments, data

mapping and phased implementation. *Interoperability testing* is critical to ensure that the model interacts effectively with different cybersecurity tools and systems. Developing a *scalable architecture* allows for customization and expansion as needed, while *continuous monitoring and feedback loops* help track performance and resolve issues promptly (Saha et al., 2024). By addressing these risks with strategic planning and robust mitigation strategies, organizations can successfully integrate language models into their cybersecurity infrastructure, enhancing rather than disrupting existing workflows and improving overall security.

- Model Transparency:** For the application of language models in the cybersecurity of organizations, it is not only important to understand the internal mechanism of these models, but also to understand how these mechanisms interact with and influence human decision-making processes (Handler, Larsen & Hackathorn, 2024). Transparency is critical to build trust in the model's outputs especially in situations where the stakes are high and security decisions have far-reaching consequences. Without adequate transparency, there is a risk of misinterpreting the results of the model, over-reliance on automated decisions, and biases, that can lead to incorrect or harmful actions. Misinterpretation can lead to security teams acting on incorrect information, while over-reliance on the model can lead to reduced human oversight (Škobo & Petričević, 2023).

To effectively mitigate these risks, a socio-technical approach is required- that combines robust technical measures with careful consideration of human and organizational factors. This approach includes implementing comprehensive explanation frameworks, regular training, continuous model testing, and ensuring active human oversight. Embedding *comprehensive explanation frameworks* such as attention visualization (Vig, 2019), feature importance (de Zarzà, de Curtò, Roig & Calafate, 2023) techniques or simpler summary explanations (Liao & Vaughan, 2023) into the model's interface is crucial to make the decision-making process more transparent and understandable for users (Handler et al., 2024, and Ali et al., 2023). *Regular training sessions* for security teams are essential to ensure they can effectively interpret and critically evaluate the model's outputs. Additionally, continuous auditing of the model's logic and decision paths helps identify and correct errors, ensuring ongoing reliability and trustworthiness. Moreover, involving human oversight in critical decision-making processes ensures that automated recommendations are balanced with expert judgement (Ray, 2023).

These strategies highlight the importance of integrating technological tools with human factors to enhance the transparency, trust, and effectiveness of language models in cybersecurity operations. By maintaining this socio-technical balance, language models can significantly contribute to security operations while maintaining clarity and accountability (Dong et al., 2023; Kasneci et al., 2023; Schwartz, Yaeli & Shlomov, 2023).

- Bias:** Bias is a major challenge when language models are used in cybersecurity due to their susceptibility to data bias, algorithmic bias, and human bias (Wu, Duan, & Ni, 2023). Specific risks include perpetuating existing biases that lead to biased threat assessments, incorrect prioritization of threats and overlooking critical vulnerabilities, which can compromise overall security (Gupta et al., 2023; Kasneci et al., 2023). Moreover, biased models may unfairly target or neglect certain groups, which can lead to ethical and legal issues (Gupta et al., 2024).

Addressing these biases is not merely a technical task but a socio-technical challenge that requires a holistic approach. Mitigation strategies must include not only the technical solutions of ensuring diverse and representative training data, developing models with bias detection and correction mechanisms, but also adhering to strict ethical guidelines. Regular audits and bias reviews are essential, as they enable the identification and correction of biases that might emerge over time. Maintaining human oversight

throughout the decision-making process is equally crucial, as it ensures that the nuanced judgement, which automated systems might overlook, are properly considered (Sanclemente, 2023; Wu, Duan, & Ni, 2023).

By implementing these comprehensive strategies, organizations can improve the reliability, fairness and overall effectiveness of language models in cybersecurity. This approach ensures that assessments are not only technically sound but also unbiased and ethically responsible, allowing for more equitable responses to threats (Al-Mansoor & Salem, 2023).

- **Model Safety:** One pressing issue with customized language models for cybersecurity is model safety. This involves ensuring the robustness and reliability of language models in the face of adversarial attacks, data poisoning, adversarial prompting, and concept drift (Gupta et al., 2023; Zou, Wang, Kolter & Fredrikson, 2023). Specific risks include adversarial attacks where attackers manipulate the model through misleading inputs, data poisoning that corrupts the model's training process, adversarial prompts that trigger malicious or erroneous outputs, and concept drift where the model's performance degrades over time as the underlying data distribution changes (Schwinn, Dobre, Günnemann & Gidel, 2023). Additionally, the impact on human decision-making and trust is a significant risk. Model safety extends beyond technical robustness to include how cybersecurity professionals interact with the model and trust its outputs (Handler et al., 2024). If the model is not safe or reliable, it can lead to incorrect or dangerous decisions, eroding trust in the system. This is a socio-technical issue because it also involves understanding and managing the human response to potential model failures.

To effectively minimize these risks, practitioners must perform rigorous data validation to ensure the integrity and quality of the input data, and conduct adversarial training to simulate and prepare for potential attack scenarios (Schwinn et al., 2023). However, these technical measures alone are insufficient. Constant human oversight is essential to detect and correct anomalies that automated system might miss, highlighting the need for a balanced socio-technical approach (Gupta et al., 2024).

In addition to these technical safeguards, organizations must prioritize organizational preparedness and response by developing clear protocols, training staff, and fostering a culture of awareness and readiness to act when a model behaves unexpectedly or incorrectly. This involves aligning technology with human and organizational processes to ensure a coherent and effective response to model failures or attacks (Aksoy, 2024).

Finally, balancing automation with human oversight is crucial. While LLMs can automate many cybersecurity tasks, they must not replace human judgement, especially in high stakes situations. Understanding the limitations of both technology and the humans who interact with it is essential for creating systems that support effective collaboration between the two (Yu, Lee, Xie, Billah & Carroll, 2024). By combining these comprehensive, socio-technical strategies-including technical, organizational, and human-centered approaches- organizations can significantly improve the security, robustness and reliability of cybersecurity language models to protect against the evolving landscape of cyber threats (Zou et al., 2023).

7. Conclusion

The use of language models in cybersecurity is a promising area of research due to their ability to facilitate effective decision-making for data-centric and knowledge-intensive tasks. Organizations can achieve a more proactive and dynamic approach to cybersecurity by utilizing the vast knowledge and adaptability of language models to guarantee a secure digital ecosystem for users and businesses. This article provides a comprehensive review of different applications of language models in

cybersecurity and find that the researchers are developing customized language models for content analysis, software and system analysis, threat intelligence and monitoring, and cyber vetting. Beyond their technical capabilities, the deployment of language models necessitates a careful consideration of their socio-technical impact. Language model-based cybersecurity solutions can reshape how decisions are made and can enable more informed and data-driven decisions. However, this shift introduces new challenges, such as the need for transparency in model operations, data protection and privacy, managing biases, ensuring model safety, and addressing infrastructure requirements and integration complexity. This review also discusses the specific considerations to ensure the optimal and safe operation of the customized language models. The main recommendations include: (1) Organizations must develop strategies that balance automation with human oversight, ensuring that the introduction of language models enhances rather than disrupts existing workflows. (2) the development of comprehensive strategies for integrating language models into existing cybersecurity systems to ensure compatibility with current systems and minimize disruption. Effective integration strategies will improve the overall security posture without causing significant changes to existing workflows. (3) Implement regular updates and adaptable training schedules to keep language models current with evolving cybersecurity threats. While continuous training can be resource-intensive, it is crucial for maintaining the effectiveness of the models. To manage cost, practitioners may consider prioritizing critical updates and using incremental training techniques. (4) Use diverse training datasets and robust evaluation methods to detect and mitigate biases in language models. Ensuring that models are trained on a wide range of data can reduce algorithmic and data biases, leading to fair and accurate results in cybersecurity applications. (5) Improve model transparency with techniques such as attention visualization and feature importance to increase trust among security experts. (6) Manage the influence of language models on organizational decision-making processes to avoid over-reliance on automated outputs and ensure that human judgement remains central to critical security operations. (7) Encourage collaboration between cybersecurity experts and AI researchers to develop language models that are closely aligned with the specific needs and challenges of the cybersecurity domain.

CRedit authorship contribution statement

Ramanpreet Kaur: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Tomaž Klopučar:** Writing – review & editing, Funding acquisition. **Dušan Gabrijelčić:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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